Machine Learning Capstone Project

California Housing Dataset

Regression • Classification • Clustering

Dataset size: 20,640 records × 10 columns

Prepared from the supplied housing.csv and the provided Capstone Project Manual.

I. Introduction

The goal of this project is to demonstrate a reproducible machine-learning workflow on the California Housing dataset. The work includes exploratory data analysis, preprocessing, one regression model, one binary classification model, and unsupervised clustering. The dataset contains geographic, housing, population, income, and ocean-proximity features. The continuous target for regression is **median_house_value**.

Dataset overview

Item	Description
Rows / columns	20,640 / 10
Regression target	median_house_value
Numeric predictor features	8
Categorical feature	ocean_proximity
Missing values	207 in total_bedrooms
Target median	$179,700
Target mean	$206,856

II. Data Preprocessing & EDA

The raw dataset has 20,640 rows and 10 columns. There are 207 missing values, all in total_bedrooms. The modeling pipelines impute numeric missing values with the median and categorical missing values with the most frequent category. Numeric features are standardized and ocean_proximity is one-hot encoded. A fixed random seed (42) is used for reproducibility. The train-test split is 80% training and 20% testing for the supervised tasks.

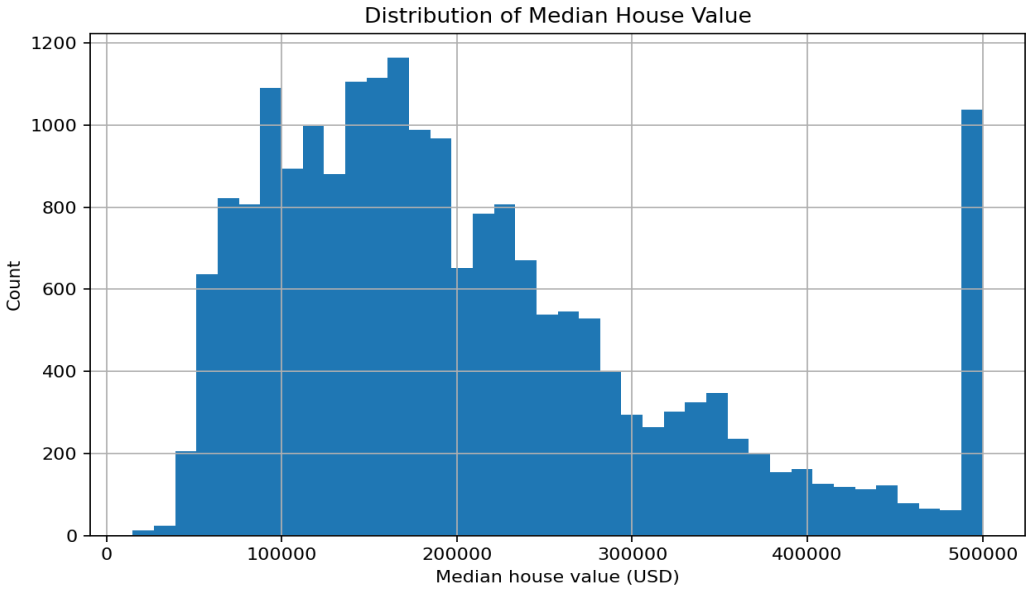


Figure 1. Distribution of the median house value target.

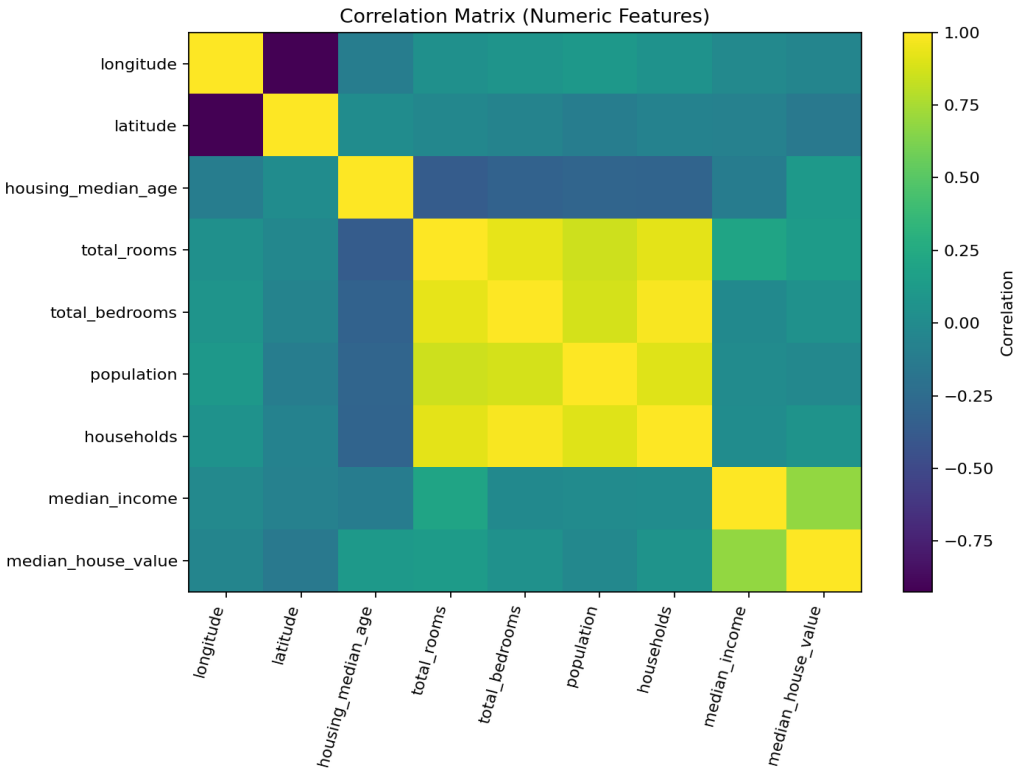


Figure 2. Correlation matrix for numeric columns.

III. Modeling and Results

A. Regression — Linear Regression

Linear Regression predicts the continuous median house value. On the held-out test set, MAE = $50,670.49, MSE = 4,908,290,571, RMSE = $70,059.19, and R^2 = 0.625. An R^2 of 0.625 means the fitted model accounts for about 62.5% of the variance in test-set target values. Errors remain substantial in dollar terms, so predictions should be treated as estimates rather than appraisals.

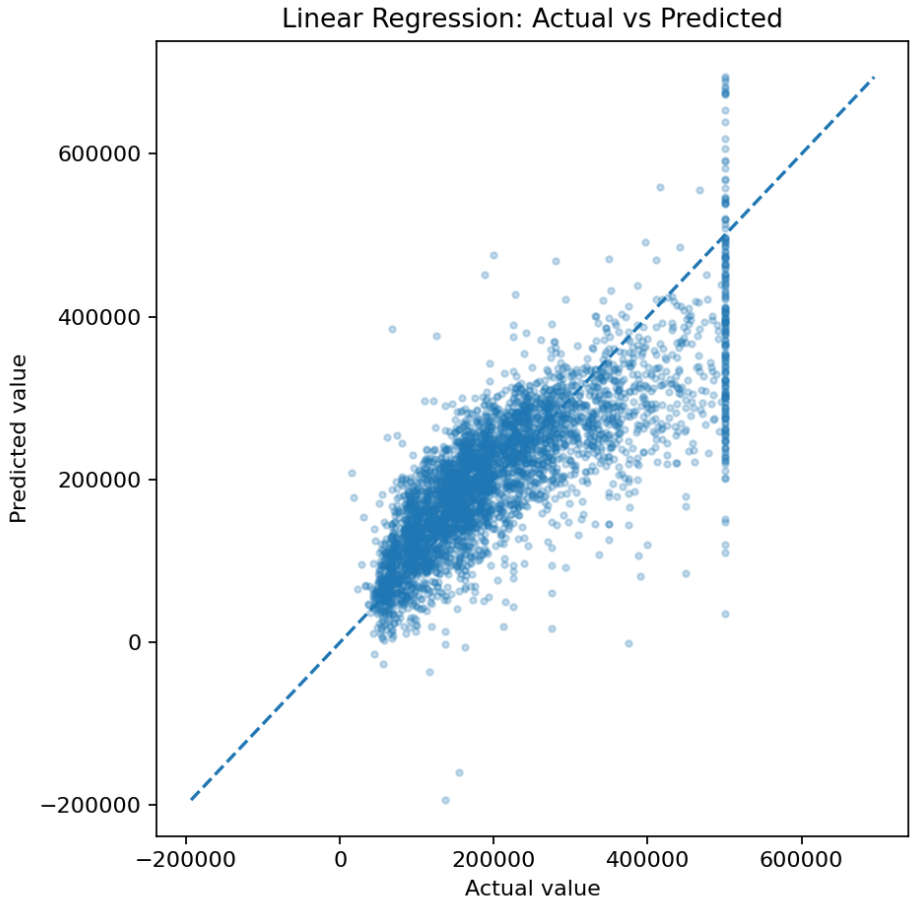


Figure 3. Actual versus predicted values; the dashed diagonal represents perfect predictions.

B. Classification — Logistic Regression

The supplied dataset has no original binary classification target. For this required demonstration, a derived label is used: class 1 means median_house_value is above the dataset median ($179,700); class 0 means it is at or below that threshold. Logistic Regression achieved accuracy 0.838, precision for class 1 0.834, recall for class 1 0.844, and F1 0.839. Confusion matrix rows are actual classes [0, 1] and columns are predicted classes [0, 1].



Confusion matrix: TN=1718, FP=347, FN=322, TP=1741. Precision measures how many predicted class-1 cases are truly class 1; recall measures how many actual class-1 cases are found.

C. Clustering — K-Means

K-Means was applied to the predictor features, excluding median_house_value and the derived classification label. Candidate values K=2 through K=8 were assessed using inertia (Elbow Method) and a sampled silhouette score. The highest sampled silhouette among these candidates occurred at K=2, with score 0.408 calculated on 2,500 reproducibly sampled rows. The final cluster sizes were cluster 0: 18,277, cluster 1: 2,363. This choice is based on the tested range and silhouette criterion; it is not proof that two clusters are the only meaningful structure.

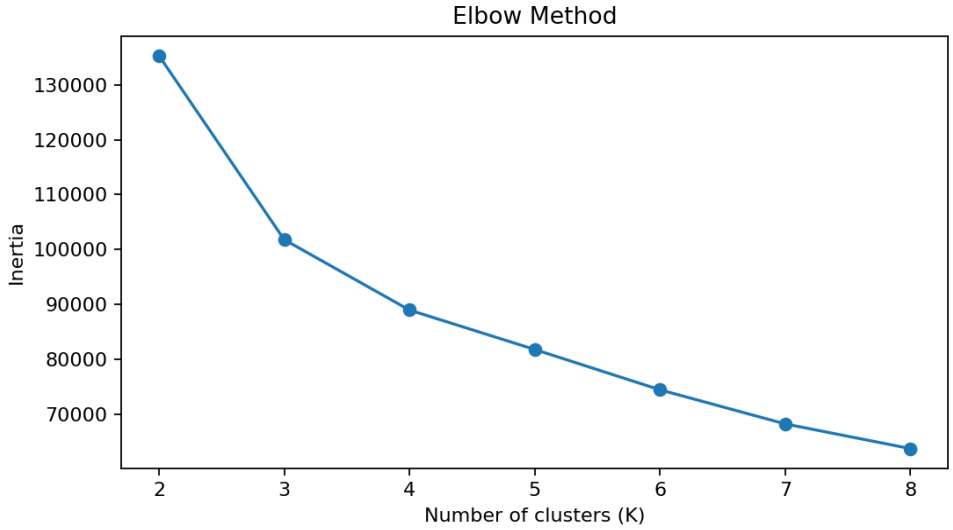


Figure 4. Inertia across candidate K values.

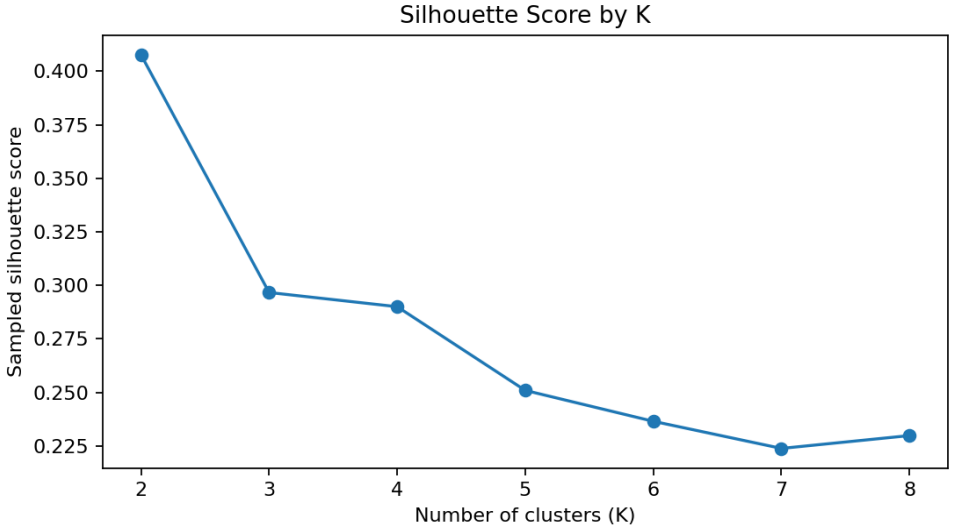


Figure 5. Sampled silhouette score across candidate K values.

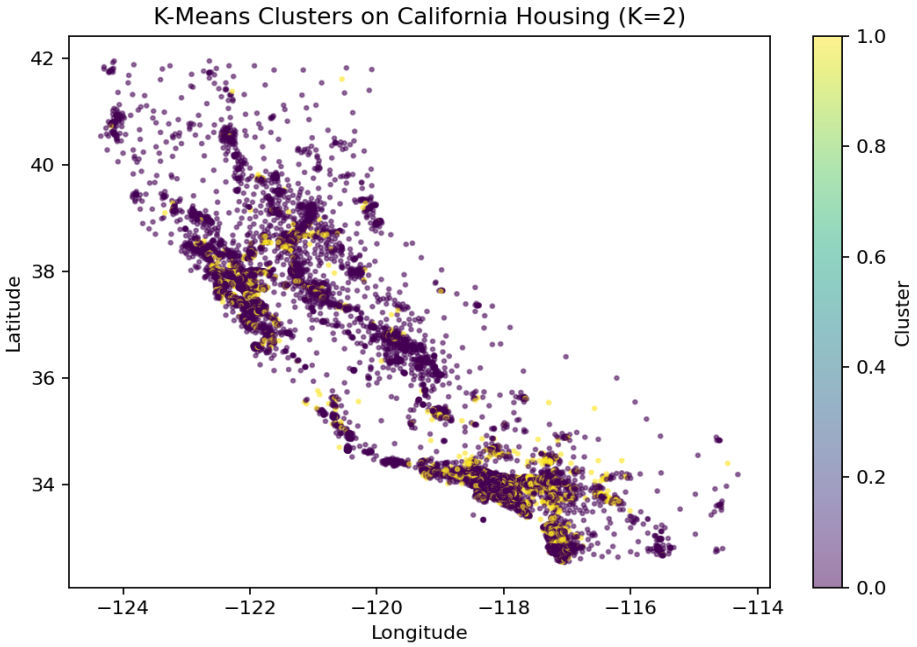


Figure 6. Geographic visualization of cluster assignments using longitude and latitude.

IV. Conclusion

The project completed all three required machine-learning task types. Linear Regression achieved R^2=0.625 and RMSE=$70,059. Logistic Regression achieved 83.8% accuracy for the derived above/below-median classification task. K-Means selected K=2 among K=2–8 using the highest sampled silhouette score (0.408). Results are reproducible using the supplied script and fixed random seed.

Challenges and next steps

The target value is capped at $500,001 for some records, and a linear model may not capture nonlinear relationships. Future work could compare Random Forest or Gradient Boosting regressors, tune hyperparameters with cross-validation, and evaluate subgroup errors. For classification, the derived label should be clearly identified as a demonstration label; a genuine classification project would use an independently meaningful binary target. For clustering, alternative K values and feature subsets should be interpreted alongside domain context.

Reproducibility and submission

Code: run_project.py. Notebook: notebooks/California_Housing_Capstone.ipynb. Metrics and figures: results/. The repository should be uploaded to a public GitHub repository named ML-Capstone-Project. The required Google Colab notebook link must be inserted below and shared with the grading team before final submission.

Google Colab notebook link: [PASTE YOUR SHAREABLE COLAB URL HERE]

Data source note: This report analyzes the user-provided housing.csv file. The dataset name and column schema are consistent with the commonly distributed California Housing dataset; the file itself was supplied with the project request.